MILITARY SPECIFICATION

SWITCHBOARD, TELEPHONE MANUAL SB-22( )/PT

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of small light weight portable, immersion proof, retractable cord type field telephone switchboard, designated as Switchboard Telephone Manual SB-22( )/PT, Switchboard Accessory Kit MX-2915( )/PT and the following subassemblies of SB-22( )/PT (see 6.4):

Telephone Circuit, Line Jack TA-222( )/PT
Telephone Circuit, Trunk Jack TA-326( )/PT
Telephone Circuit, Operators TA-221( )/PT

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder, Lead Alloy, Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Forms

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, U.S. Army Communications-Electronics Command and Fort Monmouth, Fort Monmouth, New Jersey 07703, Attention: DRSEL-~~PED-CM~~, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Inspection Terms and Definition
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

SIGNAL CORPS

SC-C-33721	Jack Assembly (Trunk)
SC-B-33731	Jack Requirements
SC-D-35754	Reel and Plug Assembly
SC-C-35755	Switchboard Cord Assembly
SC-C-35756	Cord Assembly
SC-DL-84313	Case Assembly SB-22( )/PT
SC-C-84439	Magneto Signal Assembly
SC-C-84478	Jack Assembly (Line)
SC-DL-84634	Switchboard Telephone Manual SB-22( )/PT
SC-B-140974	Electrical and Mechanical Adjustments for Jacks
SC-DL-147776	Switchboard Accessory Kit MX-2915( )/PT

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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2.2 Other Publication. Army, SPI 1G00012 Special Packaging Instructions.

3. REQUIREMENTS.

First Article. When specified the contractor shall furnish sample unit(s) for First Article inspection and approval (see 4.3).

3.1 Description. Switchboard SB-22( )/PT is a small portable field type telephone switchboard equipped with 12 individual plug in line units. Connection between lines are made with plugs and cords. The cords are of the retractable type. Included with each switchboard is one MX-2915( )/PT, an Accessory Kit containing subassemblies to increase the line capacity when the switchboard is operated in tandem with another SB-22( )/PT switchboard. A subassembly to provide a common battery trunk, replacing one line unit is provided. One Headset-Microphone H-182( )/PT is provided with each SB-22( )/PT.

3.2 Construction. The Switchboard SB-22( )/PT, Telephone Circuit Line Jack, TA-222( )/PT, Telephone Circuit, Trunk Jack, TA-326( )/PT and Telephone Circuit, Operator's TA-221( )/PT shall be constructed in accordance with Drawing SC-DL-84634, the Accessory Kit MX-2915( )/PT shall be constructed in accordance with SC-DL-147776 and the Headset-Microphone H-182( )/PT shall be constructed in accordance with specification MIL-H-55460.

3.3 Parts, materials, and processes: general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.3.1 Soldering and Brazing.

3.3.1.1 Brazing. Brazing of steels, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883.

3.3.1.2 Soldering.

3.3.1.2.1 Solder. Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes; and composition SN62 or SN63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.3.1.2.2 Acid or acid salts. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

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3.3.1.2.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not become loosened.

3.4 Marking.

3.4.1 Marking. Marking shall conform to Specification MIL-M-13231 (see 4.4). Front panel marking shall be Group 1 as described in that specification.

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial Numbers. Switchboard SB-22( )/PT shall be serial numbered in accordance with a block of serial numbers provided by the contracting officer.

3.5 Telephone Circuit, Line Jack TA-222( )/PT and Telephone Circuit Trunk Jack TA-326( )/PT.

3.5.1 The magneto signal shall conform to the construction and inspection requirements of Drawing SC-C-84439.

3.5.2 The telephone plug and cord assembly shall be constructed in accordance with Signal Corps Drawing SC-C-35755.

3.5.2.1 The cord shall be constructed in accordance with Signal Corps Drawing SC-D-35756.

3.5.3 Cord Reels. The cord reel assembly shall be constructed in accordance with Signal Corps Drawing SC-D-35754.

3.5.3.1 The cord reel assembly shall withstand, without damage, 100,000 consecutive operations under normal operating conditions (see 4.9.3).

3.5.4 Telephone Jacks. The telephone jacks used in Telephone Circuit Line Jack TA-222( )/PT shall be constructed in accordance with Drawing SC-C-84478 and shall meet the electrical and adjustment requirements of Drawing SC-D-140974. The telephone jacks used in Telephone Circuit Trunk

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Jack TA-326( )/PT shall be constructed in accordance with Drawing SC-C-33721 and shall meet the electrical and adjustment requirements of Drawing SC-B-33731.

3.5.5 Signal, Non-Operate, Shock. The signal shall not operate when deenergized and subjected to the mechanical shock specified in 4.9.1.

3.5.6 Operational.

(a) Both the line and trunk jack magneto signals shall operate from black to white, closing the circuit of its connector receptacle terminals 1 to 3, when a 30 volt 20 Hz signal is applied to contacts 2 and 4 of its connector receptacle from a 1000 ohm impedance generator (see 4.9.2(a)).

(b) Insertion of a PJ051 plug into the line or trunk jack shall operate its magneto signal from white to black, opening the circuit of its connector receptacle terminals 1 to 3 (see 4.9.2(b)).

(c) A 0 dBm, 300 Hz signal shall be transmitted from across the Line Jack connector receptacle contacts 2 and 4 to the Tip Ring contacts with a loss not greater than 0.2 dB (see 4.9.2(c)).

(d) A 300 Hz, 0 dBm signal shall be transmitted from across the Trunk Jack connector receptacle contacts 2 and 4 to the Tip and Ring contacts with a loss not greater than 2.0 dB (see 4.9.2(c)).

3.6 Telephone Circuit, Operators TA-221( )/PT.

3.6.1 Buzzer.

(a) The buzzer shall operate, producing an audible sound when 2 volts direct current is applied (positive voltage applied to terminal 2. The buzzer shall operate and the current shall not exceed 75 milliamperes when 4 volts direct current is applied (see 4.10.1).

(b) The buzzer shall have an operating life of 100,000 on-off cycles consisting of 5 seconds on with 4 volts direct current applied and 5 seconds off (see 4.10.1.1).

3.6.2 Telephone Lever Key. The telephone lever key shall have the following characteristics:

(a) Each key shall be capable of 180,000 operations without excessive wear or damage to any part thereof while alternately opening and closing a circuit through which one-half ampere of direct current, is flowing (see 4.10.2.1).

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(b) The potential drop across any pair of closed contacts, measured at the point where external leads are connected, shall not exceed 5.5 millivolts, when the contacts are carrying 100 milliamperes direct current (see 4.10.2).

3.6.3 Hand Generator. The hand operated generator shall conform to Specification MIL-G-11994 Generator, Ringing, Hand G-42( )/PT.

3.6.4 Panel Lamp. The panel lamp shall be capable of 50,000 operations of the controls and switches incorporated in the lamp housing (see 4.10.4).

3.6.5 Operational.

(a) With a 3 volt dc voltage source capable of delivering at least 150 milliamperes applied to its connector receptacle P302 contacts 2 and 4, the operator's lamp cap shall light when pulled to its detent position away from the panel. The lamp shall go out when pushed to its detent position toward the panel (see 4.10.3(a)).

(b) Connecting an external jumper between contacts 1 and 3 of receptacle P302, the operator's lamp shall light when the VIS-AUD switch is set to VIS position and the buzzer shall produce an audible noise when the VIS-AUD switch is set to AUD. Neither lamp nor buzzer shall operate when the VIS-AUD switch is in its OFF position (see 4.10.3(b)).

(c) With an external 90 volt, 20 Hz, signalling supply connected between the operator's connector receptacle P305 contacts 2 and 4 and the operator's ring switch set to its PWR RING FWD position, the 20 Hz 90 volt signal shall be present across the tip and sleeve contacts of the operator's plug P301 (see 4.10.3(c)).

(d) With the external ringing supply connected as in (c) and the operator's ring switch set to its RING BACK position, the 20 Hz 90 volt signal shall be present across the tip and ring contacts of the operator's plug P301 (see 4.10.3(d)).

(e) A ringing signal shall appear across the tip and sleeve contacts of the operator's plug P301 when the operator's generator is rotated (see 4.10.3(e)).

(f) A ringing signal shall appear across the tip and ring contacts of plug P301 when the operator's generator is rotated and the lever switch S301 is set to its RING BACK position (see 4.10.3(f)).

(g) With 3 volts dc applied between receptacle P303 contacts 2 and 4, and with Handset-Microphone H-182( )/PT connected to the TA-221( )/PT with its push-to-talk switch in its operated position, the operator's

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voice at the H-182( )/PT microphone shall be heard in the earphone. Application of an audio tone signal from tip to sleeve contacts of the operator's plug shall appear at the operator's earphone (see 4.10.3(g)).

3.7 Switchboard and Battery Case. The switchboard case and cover and battery case shall be constructed in accordance with Signal Corps Drawing SC-DL-84313 and shall meet the following additional requirements.

3.7.1 Spring Binding Posts. The spring binding posts shall conform to Specification MIL-P-55149/9, Posts, Binding, Electrical (Insulated), type PB09NC01.

3.7.2 Connections, Potential Drop. The potential drop through the contact of the rear binding post to the interior pins shall not exceed 2 millivolts with 100 milliamperes flowing through the contact (see 4.11).

3.7.3 Switchboard Case, Immersion, The switchboard case shall show no evidence of leakage when tested in accordance with 4.16(a).

3.7.3.1 Battery Case Immersion. The battery case shall show no evidence of leakage when tested in accordance with 4.16(c).

3.8 Accessories.

3.8.1 Headset-Microphone. One Headset-Microphone H-182( )/PT constructed in accordance with Specification MIL-H-55460 shall be furnished with each Switchboard SB-22( )/PT.

3.8.1.1 Visual and Mechanical. The headset-microphone shall meet the requirements of MIL-STD-252 as specified in 4.26.

3.8.1.2 Operational. The completely assembled equipment shall be operationally tested for pick-up, transmission and output of a voice signal and operation of the switch in each position, when tested in accordance with 4.27.

3.8.2 Accessory Kit. One Accessory Kit MX-2915( )/PT, constructed in accordance with Signal Corps Drawing SC-DL-147776 and meeting the applicable requirements of this specification shall be furnished with each Switchboard SB-22( )/PT.

3.8.2.1 Accessory Kit, Immersion. The accessory kit case shall show no evidence of leakage when tested in accordance with 4.16(b).

3.9 Switchboard Telephone Manual SB-22( )/PT.

3.9.1 Continuity. The wiring and contacts shall show continuity within their respective circuits (see 4.12.1).

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3.9.2 Insulation Resistance. The insulation resistance of all circuits shall not be less than 100 megohms. During moisture resistance cycling, the insulation resistance shall not be less than 1 megohm at 500V DC (see4.8).

3.9.3 Dielectric Withstanding Voltage. Air gaps in jacks, switches and similar items shall have a dielectric withstanding voltage equal to or greater than 500 volts direct current, with the exception of the night alarm circuit. The night alarm circuit shall have a dielectric withstanding voltage of 100 volts direct current or peak alternating current. Exclude circuits containing capacitors C-301 and C-302 (see 4.7).

3.9.4 Crosstalk. The crosstalk coupling between any two circuits shall not be more than -60dbm at any frequency between 300 and 3500 c.p.s. with the disturbing power at 1 milliwatt level (see 4.12.2).

3.9.5 Transmission Loss. The transmission loss between pairs of line binding posts of any two Line Jack Circuits TA-222( )/PT connected through the switchboard shall not exceed 0.2 db for any frequency between 300 and 3000 cycles. The transmission loss between pairs of binding posts on the Trunk Jack TA-326( )/PT and Line Jack TA-222( )/PT connected through the switchboard shall not exceed 2.0 db for any frequency between 300 and 3000 c.p.s. (see 4.12.3).

3.9.6 Bridging Loss. The transmission loss between pairs of binding posts on Trunk Jack TA-326( )/PT and Line Jack TA-222( )/PT connected through the switchboard shall not exceed the values listed below for any frequencies between 300 to 3000 c.p.s., when Operator's Circuit TA-221( )/PT is bridged across the talking circuit (see 4.12.4).

Frequency	Loss db Max
300	4.0
600	3.0
1000	2.0
1500	2.0
2000	1.5
2500	1.5
3000	1.0

3.10 Electromagnetic Interference. The equipment shall comply with the following requirements of Table A-II-MIL-I-11748, Notice 4, MIL-STD-461A, Equipment Class III a. Test Method RE02 (see 4.13).

3.11 Operational Requirements. The switchboard shall be capable of answering and completing connections for the following classes of services:

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Between local battery field telephones.
Between field telephones and trunks.
Conference calls between field telephones and trunks.
Remote control radio circuits.
Voice frequency teletypewriter circuits.
Between local battery field telephones and common battery trunk when Telephone Circuit, Trunk Jack TA-236( )/PT is used and shall be capable of operation in tandem with one or more switchboard, Telephone, Manual SB-22( )/PT (see 4.12.5)

3.12 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (see 4.14).

3.13 Service Conditions. The equipment shall meet the following service conditions:

3.13.1 Equipment Operating. Any of the following conditions, successively or in combinations likely to be encountered during worldwide operations:

(a) Temperature. Ambient temperature in the range of +150°F to -65°F. (The 150°F temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at one time (see 4.18).

(b) Relative Humidity. Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours (see 4.17)

(c) Elevation. Elevation up to 10,000 feet above sea level (see 4.15).

3.13.2 Equipment Nonoperating. Any of the following conditions, successively or in combinations encountered during worldwide short-term storage and transit:

(a) Temperature. Exposure in the range of +160°F to -80°F; however, exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time (see 4.18).

(b) Relative Humidity. Relative humidity up to 97 percent for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours (see 4.17).

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(c) Elevation. Elevation up to 50,000 feet above sea level (see 4.15).

(d) Immersion. Three feet of water for 2 hours (see 4.16).

(e) Vibration, Bounce, and Shock. As described in 3.13.

3.14 Vibration, Bounce and Shock.

3.14.1 Vibration. The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not sustain any physical damage (see 4.19).

3.14.2 Bounce. The equipment shall be capable of meeting the visual and mechanical inspection, as specified in 4.25; the Operational Test, specified in 4.12.5(a) through 4.12.5(l) and show no physical damage except surface abrasion after subjection to the bounce test specified in 4.20.

3.14.3 Shock Bench Handling. The equipment shall withstand the test specified in 4.21 without physical damage or deterioration of operating efficiency.

3.15 Drop Test. A complete SB-22 case, less line and trunk packs, operator's pack and battery case shall display no damage except for slight abrasions and shall meet the immersion requirement of paragraph 3.7.3, following the drop test of paragraph 4.22.

3.16 Specular Gloss. The 60 degree specular gloss of the SB-22( )/PT case shall be less than 30 (see 4.23).

3.17 Preconditioning. The equipment shall be capable of meeting the requirements herein without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.18 Technical Literature, Tools, and Running Spare Parts. Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2).

3.19 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.14).

3.20 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the following paragraphs herein (see 4.26).

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Construction	Para 3.2
Soldering and Brazing	3.3.1
Marking	3.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.3). Does not include packaging.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.24).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor on units designated in 4.3.1.

4.3.1 First Article Units. The contractor shall furnish two First Article units of Switchboard, Telephone, Manual SB-22( )/PT and the following, unless otherwise specified:

3 each Generator, Ringing, Hand G-42( )/PT
3 each Magneto Signal
2 each Panel light
2 each Buzzer
2 each Cord Reel
2 each Lever Key
6 each Posts, Binding Electrical

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4.3.2 First Article Inspection. The First Article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for Group A, Group B and Group C (see Tables I, II, III respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units; and Group C as specified in Table III. The first article inspection of Switchboard, Telephone Manual SB-22( )/PT shall be performed in the following order: (1) vibration, (2) bounce, (3) shock, bench handling, (4) immersion. Other first article environmental tests may precede, follow or be interspersed with the foregoing. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.3.3 First Article Data. The First Article test plan and test report shall be furnished as required in the contract or purchase order.

4.3.4 Failures. One or more failures may be cause for refusal to grant first article approval.

4.4 Inspections Covered by Subsidiary Documents: The following shall be inspected under the applicable subsidiary document as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Marking	3.4
Magneto Signal	3.5.1
Generator, Ringing, Hand G-42( )/PT	3.6.3
Post, Binding, Electrical	3.7.1

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. The contractor shall perform the inspection specified in 4.4, 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling, inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLS indicated in Table I.

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TABLE I - GROUP A INSPECTION OF EQUIPMENT

INSPECTION	Req Para	Insp Para	AQL Major	AQL Minor
VISUAL AND MECHANICAL	3.20	4.26		
Telephone Circuit, Line Jack TA-222( )/PT	3.2	4.26	1.0 DPHU	4.0 DPHU
Telephone Circuit, Trunk Jack TA-326( )/PT	3.2	4.26	1.0 DPHU	4.0 DPHU
Telephone Circuit, Operators TA-221( )/PT	3.2	4.26	1.0 DPHU	4.0 DPHU
Switchboard Case	3.7	4.26	1.0%	4.0%
Battery Case	3.7	4.26	1.0%	4.0%
Accessory Kit MX-2915( )/PT	3.8.2	4.26	1.0%	4.0%
Switchboard, Telephone Manual SB-22( )/PT	3.2	4.26	2.5 DPHU	10.0 DPHU
Headset-Microphone H-182( )/PT	3.8.1.1	4.26	1.0 DPHU	4.0 DPHU
Telephone Circuit, Line Jack TA-222( )/PT and Telephone Circuit, Trunk Jack TA-326( )/PT				
Dielectric withstanding voltage	3.9.3	4.7)	1.0% for the Group combined	
Insulation resistance	3.9.2	4.8)		
Signal, non operate, shock	3.5.5	4.9.1	1.0%	
Operational 2/	3.5.6	4.9.2	1.0%	
Telephone Circuit,Operators TA-221( )/PT				
Dielectric withstanding voltage	3.9.3	4.7	1.0%	
Insulation resistance	3.9.2	4.8	1.0%	
Buzzer 1/	3.6.1	(a)4.10.1	1.0%	
Telephone lever key contact resistance 1/	3.6.2	(b)4.10.2	1.0%	
Operational 2/	3.6.5	4.10.3	1.0%	
Switchboard Case				
Connections, potential drop	3.7.2	4.11	1.0%	
Switchboard Telephone Manual SB-22( )/PT				
Continuity	3.9.1	4.12.1)		
Dielectric withstanding voltage	3.9.3	4.7)		

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Insulation resistance	3.9.2	4.8)	
Transmission loss	3.9.5	4.12.3)	6.5 DPHU for the
Crosstalk	3.9.4	4.12.2)	Group combined
Bridging loss	3.9.6	4.12.4)	
Operational	3.11	4.12.5)	1.0%
Headset-Microphone H-182( )/PT			
Operational	3.8.1.2	4.27	0 DPHU

DPHU= Defects per 100 units

1/ - To be performed oncomponent lots prior to assembly and wiring into TA-221( )/PT.

2/ - To be performed on assembly lots prior to mounting into switchboard.

4.5.2 Group B Inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105.

4.5.2.1 Group B Sampling Plans. The Group B sampling plans for the AQLs listed in Table II shall be Inspection Level S-4.

4.5.2.2 Order of Inspection Within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE II - GROUP B INSPECTION OF EQUIPMENT

INSPECTION	Req Para	Insp Para	AQL
IMMERSION			
Switchboard case	3.7.3	4.16(a)	2.5%
Battery Case	3.7.3.1	4.16(c)	2.5%
Accessory Kit MX-2915( )/PT	3.8.2.1	4.16(b)	2.5%
INTERCHANGEABILITY	3.12,3.19	4.14	2.5%

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. Samples shall be selected in accordance with 4.5.3.1.

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4.5.3.1 Sampling for Group C Inspection. Two SB-22A/PT for each Group C inspection shall be selected at random from every 500 units or fraction thereof produced or at intervals of 3 months whichever occurs first. The first samples selected shall be at the start of the contract, from the first quality conformance inspection production lot.

TABLE III GROUP C INSPECTION

Inspection	Req Para	Insp Para
LIFE TEST		
Cord Reel	3.5.3.1	4.9.3
Buzzer	3.6.1(b)	4.10.1.1
Telephone Lever Key	3.6.2(a)	4.10.2.1
Panel Lamp	3.6.4	4.10.4
Electromagnetic Interference	3.10	4.13
Elevation	3.13.1(c),3.13.2(c)	4.15
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NOTE: Subassemblies and parts subjected to the life tests of Group C inspection shall not be used in assembly of equipment or delivered on orders.

4.5.4 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Disposition of Nonconforming Product. Disposition of nonconforming product shall conform to Standard MIL-STD-105. Such product shall be suitably tagged or identified to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

4.6 Bounce Preconditioning. The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection (see 3.17).

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4.7 Dielectric withstanding voltage. The insulation between all circuits and between circuits and metal frames shall be tested for dielectric strength in accordance with the requirements of method 301 of MIL-STD-202 (see 3.9.3).

4.8 Insulation resistance. The insulation between all circuits and between all circuits and metal frames shall be tested in accordance with method 302 of MIL-STD-202 (see 3.9.2).

4.9 Telephone Circuit Line Jack TA-222( )/PT and Telephone Circuit Trunk Jack TA-326( )/PT.

4.9.1 Signal non-operate, shock. With the switchboard or the Telephone Circuit, Line Jack TA-222( )/PT or Telephone Circuit, Trunk Jack TA-326( )/PT placed on a flat surface in a horizontal position, withdraw any plug from its seat, extending the cord eighteen inches along the surface in direct line with the unit. Release the cord, permitting it to be retracted to its normal position by its reel. Impact of the plug with the panel when released shall not cause the operation of any line supervisory signal (see 3.5.5).

4.9.2 Operational, Telephone Circuit Line Jack TA-222( )/PT and Telephone Circuit Trunk Jack TA-326( )/PT.

(a) A 30 volt, 20 Hz ringing generator having a source impedance of 1000 ohms shall be applied across its connector receptacle terminals 2 and 4. An ohmmeter or other suitable means of continuity measurement shall be used to monitor contact closure across connector receptacle terminals 1 and 3 (see 3.5.6(a)).

(b) The Line or Trunk Jack magneto signal shall be set to its black position by inserting a PJ051 plug into its jack and then removing the plug (see 3.5.6(b)).

(c) A 0 dBm, 300 Hz voltage shall be applied across the connector receptacle contacts 2 and 4 while monitoring the voltage across both the tip and ring contacts of the jack and plug with a 600 ohm input dB meter (see 3.5.6(c) and 3.5.6(d)).

4.9.3 Cord reel test. The cord and plug installed in a TA-222( )/PT shall be extended and retracted its full length a total of 100,000 cycles. During the cycle, the plug shall be extended and retracted at a constant rate of 20 to 25 cycles per minute. The plug shall not strike the face of the switchboard but shall seat at the end of each cycle. At the conclusion of this test, the cord and reel assembly shall meet the requirement of 3.5.3.1.

4.10 Telephone Circuit, Operators TA-221( )/PT.

4.10.1 Buzzer

(a) The buzzer shall be connected to a 2 volt dc supply (positive voltage on terminal 2) while observing the buzzer operation (see 3.6.1(a)).

(b) The buzzer shall be connected to a 4 volt dc supply (positive voltage on terminal 2) with a milliammeter connected in series with one line. The buzzer operation shall be observed and the current shall not exceed that specified in 3.6.1(a).

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4.10.1.1 Buzzer life.

(a) The buzzer shall be connected to a 4 volt dc supply that has an automatic 5 second on, 5 second off cycle with an electrically operated counter actuated by the buzzer current.

(b) The buzzer shall be actuated for 100,000 on-off cycles to determine compliance with 3.6.1(b).

4.10.2 Telephone lever key, contact test. The resistance of the key contacts may be measured directly using a Kelvin bridge, or indirectly using the voltmeter - ammeter method, ammeter-potentiometer method or other equally sensitive means. The maximum allowable measurement error shall be 5 percent. The point of measurement shall be the point at which the external leads are normally connected to the key springs. Connections between the key springs and the measuring apparatus shall be made as specified using suitable connecting jigs or clamps, where required. With 100 milliamperes of current flowing through the contact, the potential drop across the contact shall be measured. The contact may be opened and closed 3 times to clear the connection, then a measurement shall be made, the contacts shall be opened and closed and the measurement repeated (see 3.6.2(b)).

4.10.2.1 Telephone lever key, life. The telephone lever key shall be tested as follows:

(a) Connect all of the normally closed lever spring contacts in series.

(b) Connect all of the normally open lever spring contacts in series.

(c) Connect the two sets of leads from each of the above in parallel across a 500 milliampere current regulated dc supply with an electrically actuated counter activated by the supply current.

(d) The key shall then be operated to its extreme positions in both directions of operation one hundred eighty thousand (180,000) times (see 3.6.2(a)).

4.10.3 Operational, Telephone Circuit, Operators TA-221( )/PT

(a) Apply a 3 volt dc voltage to its connector receptacle P302 contacts 2 and 4. The cap of the operator's lamp shall be pulled to its detent position away from the panel and then pushed to its detent position toward the panel while observing the operation of the lamp (see 3.6.5(a)).

(b) Connect an external wire jumper between contacts 1 and 3 of connector receptacle P302 and push the cap of the operator's lamp to its detent position toward the front panel. The VIS-AUD switch shall be set to each of its VIS, AUD and OFF positions while observing the condition of the operator's lamp and buzzer (see 3.6.5(b)).

(c) Connect an external 90 volt, 20 Hz signalling supply between contacts 2 and 4 of the operator's connector receptacle P305 and an ac voltmeter or other voltage indicator across the tip and sleeve contacts of the operator's plug P301. The operator's ring switch shall be set to its PWR RING FWD position while observing the voltage indicator (see 3.6.5(C)).

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(d) Procedure (c) above shall be repeated with the operator's ring switch set to its RING BACK position and the voltage indicator shall monitor the top to ring contacts of the operator's plug P301 (see 3.6.5(d)).

(e) Connect an ac voltmeter or voltage indicator across the tip and sleeve contacts of the operator's plug P301. The operator's generator shall be rotated while observing the voltage indicator (see 3.6.5(e)).

(f) Connect an ac voltmeter or voltage indicator across the tip and ring contacts of the operator's plug P301. The operator's generator shall be rotated and the lever switch S301 set to its RING BACK position while observing the voltage indicator (see 3.6.5(f)).

(g) Connect a Handset-Microphone H-182( )/PT, with its push-to-talk switch operated, to the operator's receptacle and a 3 volt dc supply between contacts 2 and 4 of connector receptacle P303. Connect an audio oscillator, set for 0 dBm output, across the tip to sleeve contacts of the operator's plug. Speak into the H-182( )/PT microhpone and then apply the signal from the audio oscillator while listening for the tone in the H-182( )/PT earphone (see 3.6.5(g)).

4.10.4 Panel lamp, life. The panel lamp shall be connected in a circuit with a 3 volt power source. The circuit shall be closed and opened 50,000 times where the closed period shall be of sufficient time to permit the lamp to reach full brilliance and the open period shall be the same length of time as the closed period. An electrically actuated counter shall be activated by the lamp current (see 3.6.4).

4.11 Connections, potential drop. The voltage drop through the contact of the rear binding post to the interior pins shall be measured with a current of 100 milliamperes flowing through the connection. A Kelvin bridge or equivalent ammeter-potentiometer or other approved method may be used. The measurement shall comply with 3.7.2.

4.12 Switchboard, Telephone Manual SB-22( )/PT.

4.12.1 Continuity. Except where continuity of the wiring and other connections cannot be verified by the performance of operational tests, the continuity shall be verified by a point-to-point test (see 3.9.1).

4.12.2 Crosstalk. A variable oscillator with 600 ohms output impedance shall be connected in turn to a pair of each of the line binding posts corresponding to the Telephone Circuit, Line TA-222( )/PT and Telephone Circuit, Trunk TA-326 ( )/PT. A transmission measuring set TS-559( )/FT, Daven 34-C noise measuring set or approved equivalent shall be connected to a pair of adjacent line binding posts and set for 1 milliwatt output level. Measurement results shall conform to 3.9.4.

4.12.3 Transmission loss. The transmission loss measurement shall be made by connecting a variable oscillator with 600 ohms output impedance and adjusted to deliver 1 milliwatt, to one pair of line binding posts. The associated line plug shall be inserted in the jack of each of the other line or trunk packs in turn, while measuring the output at the associated pair of binding posts with a 600 ohm input impedance dB meter. Measurement results shall conform to 3.9.5.

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4.12.4 Bridging loss. The transmission loss caused by bridging the operators circuit across two circuits connected through the switchboard shall be measured as follows:

(a) Connect a variable oscillator with 600 ohms output impedance to one pair of line binding posts of a Line Jack TA-222( )/PT.

(b) Insert the associated line plug into the jack of a Trunk Jack TA-326( )/PT and connect a dB meter, with 600 ohms input impedance, to the binding posts of the TA-326( )/PT line.

(c) Adjust the oscillator output to obtain a reading of 0 dBm on the meter for the test frequency.

(d) Connect a Handset-Microphone H-182( )/PT, modified by replacing its receiver element with a 250 ohm mominductive resistor and transmitter open at its switch, to the operator's TA-221( )/PT.

(e) Insert the operator's plug into the Line Jack TA-222( )/PT jack and measure the resulting change in reading of the dB meter. The resulting measurement shall conform to 3.9.6.

4.12.5 Operational. The basic Switchboard Telephone Manual SB-22( )/PT, containing 12- Telephone Circuit Line Jack TA-222( )/PT and 1- Telephone Circuit Operators TA-221( )/PT shall be tested when connected to another test Switchboard, Telephone Manual SB-22( )/PT containing at least 2 - Telephone Circuit Line Jack TA-222( )/PT and 1 - Telephone Circuit Trunk Jack TA-326( )/PT as follows (see 3.11):

(a) Connect a Telephone Set TA-312( )/PT to one pair of line binding posts and a second TA-312( )/PT to another pair of line binding posts. Connect the Headset-Microphone H-182( )/PT to the operators TA-221( )/PT. Insert the required batteries in the Switchboard battery holder and the holder in its contact clips.

(b) Set all magneto signals to black condition by inserting the operators plug into each of their jacks. Set the operator's lamp cap to its full position and the operator's VIS-AUD switch to VIS position.

(c) Rotate one of the TA-312( )/PT generators applying a ringing voltage to a pair of line binding posts. The corresponding TA-222( )/PT magneto signal shall operate from black to white and the operator's night alarm lamp shall light.

(d) Set the operator's VIS-AUD switch to its AUD position. The operator's buzzer shall produce an audible sound.

(e) Insert the operator's plug into the calling party's jack. The corresponding magneto signal shall operate from white to black and the operator's buzzer shall go off.

(f) Communication shall be established between the switchboard operator and the TA-312( )/PT operator.

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(g) Insert the calling party's plug into the called party's jack (connected to the second TA-312( )/PT).

(h) Rotate the operator's generator. The called party's TA-312( )/PT buzzer shall sound and a three-way communication shall be established between the switchboard operator and the two TA-312( )/PT operators.

(i) Set the operator's ring key to "RING-BACK" position and rotate the operator's generator. The calling party's TA-312( )/PT buzzer shall sound.

(j) Remove the operator's plug and rotate one of the TA-312( )/PT generators. The magneto signal corresponding to the calling party's line shall operate black to white and the operator's buzzer shall sound.

(k) Insert the operator's plug into the calling party's jack and then remove the operator's plug and the calling party's plug from their respective jacks. The operator's buzzer shall go off.

(l) Repeat the above steps - (a) through (k) for the balance of the TA-222 ( )/PT lines, including a calling party line of the switchboard under test and a called party on the test switchboard. Note that "RING BACK" through a TA-326( )/PT Telephone Circuit Trunk Jack is not possible. Only the "calling" party may "RING IN."

4.13 Electromagnetic Interference. The equipment shall be tested for compliance with para. 3.10 using the test methods and procedures of Notice 3, MIL-STD-462.

4.14 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.12 and 3.19. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts of assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.15 Elevation test. The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment).

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(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.12.5(a) through 4.12.5(k)	Operational test

(b) Reduce the barometric pressure to 20.6± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.15(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.15(a). The equipment shall meet specified performance for these measurements (see 3.13.1(c) and 3.13.2(c)).

4.16 Immersion.

(a) Switchboard SB-22( )/PT, fully equipped, with its cover locked in place as for field transportation, shall be raised in temperature 40° F or more above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within plus or minus 2°F of its initial temperature for the duration of the test, or other means to maintain water temperature may be used. The switchboard shall be removed from the temperature chamber and immediately immersed in the water tank to a minimum depth of 3 feet above the height of the equipment and held at that depth for 2 hours. After completion of the 2 hour period the equipment shall be removed and its exterior surfaces wiped dry. The case shall be opened and examined for water leakage. (see 3.7.3).

(b) Accessory Kit. - The Accessory Kit MX-2915( )/PT fully equipped and with its cover locked in place, shall be tested by the same procedure described in 4.15(a) (see 3.8.2.1).

(c) Battery case assembly. - The battery case consists of two sections each equipped with a watertight end cap. With one end cap in place and the other end cap replaced with a fixture, equipped with a valve, a pressure of 5 pounds of air shall be applied to the case. The case shall be immersed in water and any leakage points will be determined by bubbles in the water. The other end of the case and cap shall be tested in like manner. (see 3.7.3.1).

4.17 Moisture resistance test for equipment. The equipment shall be tested as follows:

4.17.1 Test conditions.

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

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(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles on the case, operators assembly and line and trunk assemblies removed from the case.

4.17.2 Test method. The equipment shall be tested in accordance with MIL-STD-810, procedure II. The measurements as specified in 4.17.2(a) shall be made prior to humidity cycling in step 4 and following conditioning in step 9. The measurements as specified in 4.17.3(b) shall be made during humidity cycling.

4.17.3 Performance.

(a) The equipment shall meet the requirements of 3.9.2 Insulation Resistance and 3.9.4 Crosstalk.

(b) The equipment shall meet the requirements of 3.9.2 Insulation Resistance.

4.17.4 Failure. If any equipment fails to meet the performance specified in 4.16.3 during cycling, it shall be realigned or readjusted once. If the equipment then fails to meet 4.17.3(b) or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.17.3(a), it does not pass the test (see 3.13.1(b) and 3.13.2(b)).

4.18 Temperature. The equipment shall be subjected to the High Temperature test of MIL-STD-810, Method 501, Procedure 1 and the Low Temperature test of MIL-STD-810, Method 502, Procedure I. The temperature and test time requirements shall conform to 3.13.1(a) and 3.13.2(a). Measurements shall be taken as follows prior to temperature testing, following temperature testing and at the specified operating temperature extremes:

Paragraph	Measurement	Test Results
4.12.5(a) through 4.12.5(k)	Operational	Full specifications performance.

4.19 Vibration. The equipment shall be subjected to the following vibration test:

(a) Secure the equipment directly to the vibration table that can be controlled within 10 percent of the specified amplitude.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitude of the outer case by optical means, or by other means provided that vibration of the part is not affected by the measurement.

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(e) Examine the internal structure, parts subassemblies when excessive vibration noise indicates that a part has loosened and examine the entire equipment following the vibration test (see 3.14.1).

4.20 Bounce. The equipment shall be subjected to the Bounce, Loose Cargo test of MIL-STD-810, Procedure XI, Part 2 (see 3.14.2).

4.21 Shock tests; bench-handling. The telephone Circuit Line Jack TA-222( )/PT assemblies, the Telephone Circuit Trunk Jack TA-326( )/PT assembly and the Operator's Telephone Circuit TA-221( )/PT assembly shall be removed from the case and placed in a suitable position for servicing on a solid, 2 inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occure during servicing (see 3.14.3).

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis, for a total of four drops.

(b) Repeat (a), with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.22 Drop. The case assembly, less line packs, trunk pack, operator's pack and battery case, shall be subjected to a series of 8 drops (1 per corner) from a height of 48 inches onto a floor or barrier of 2 inch plywood backed by concrete. At the completion of the drop test, the case assembly shall meet the requirements of 3.15 and 3.7.3.

4.23 Specular Gloss. The equipment case when measured, using Method 6101 of Federal Test Method Standard 141, shall meet the requirements for specular gloss of paragraph 3.16.

4.24 Acceptance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116, to determine conformance to the requirements of Section 5.

4.25 Rough handling test (preparation for delivery). When rough handling test is required by the contract (see 6.2(e), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling. Operational test requirement 3.11, tested in accordance with 14.12.5(a) and 14.12.5(b).

4.26 Visual and mechanical inspection. The equipment shall be examined for the defects listed in Standard MIL-STD-252. Screws used in the plastic case shall be torque tested to the following schedule:

Screw Size	Inspection
#4	3 in - lbs
#6	4 in - lbs
#8	9 in - lbs

Plugs and jacks shall mate without application of undue force and plugs shall be retained in the jacks when mated.

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4.27 Headset-Microphone Operation. The completely assembled and connected headset-microphone shall be connected to a test fixture (audio frequency amplifier having required gain to facilitate verification) and shall meet the requirements in 3.8.1.2 when tested as follows:

Step 1 - Connect headset-microphone to test fixture by means of Plug U-161/U.

Step 2 - Switch in Position 1, talk into microphone and listen at earphone. No speech should be heard. Apply a signal to the earphones. The signal should be heard.

Step 3 - Switch in Position 2, talk into microphone and listen at earphone. Speech should be heard in earphone.

Step 4 - Switch in Position 3, talk into microphone and listen at earphone. Speech should be heard in earphone.

Step 5 - Switch in Position 4, talk into microphone and listen at earphone. Speech should be heard in earphone.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI IG00012.

6. NOTES

6.1 Intended use. Switchboard Telephone Manual SB-22( )/PT is used for inter-connecting local battery telephone lines, local battery telephone lines to local battery trunks, local battery telephone lines to common battery trunks and interconnecting voice frequency telegraph lines or lines equipped for remote control radio communication.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Number of preproduction samples required.

(d) Level A or level B preservation and packing (see section 5).

(e) When rough handling test is required.

(f) Marking or shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and running spare parts required.

(j) Facsimilies required.

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6.3 Color of textiles. Standard samples of Olive Drab No. 7 may be obtained from the Commanding General, Defense Personnel Support Center, 2800 South Twentieth Street, Philadelphia, Pennsylvania, 19101. When requesting samples, indication should be given as to type of material concerned.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a latter identifying the particular design; for example: SB-22A/PT. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3).

6.6 Location of operational inspection. It is desirable that the operational inspection (4.12.5) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Definitions.

6.7.1 Inspection terms. The inspection terms used herein conform to Standard MIL-STD-109.

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.9 Provisions of this specification are the subject of international standardization agreements. When amendment, revision or cancellation of this specification is proposed the departmental custodians will inform their respective Departmental Standardization Office (DepSO) so that appropriate action may be taken respecting the international agreement concerned.

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